

- a) Explain the operation of the above circuit.
- b) Also plot the transfer characteristic and the output waveform.

EXERCISE 8

TRANSISTORS

THEORETICAL PART

The transistor is the first discrete active element discovered in the form of a semiconductor (a P and N junction). The

transistor is structurally similar to a crystal diode at its base, but contains a piece of semiconductor (P or N) above. In figure (1) we see the two basic types of transistor construction (PNP and NPN). We observe that the transistor has (3) terminals called emitter, base and collector.

In terms of behavior, NPN transistors do not differ in any way from PNP transistors. The difference is in the polarity of the bias. POLARITY: It is

the electric potential that we apply to each terminal of the transistor so that it operates in the desired way. To see the behavior of the transistor,

we will refer to our knowledge of the PN junction. In figure 2 a, we observe that two depletion regions develop along the mass of the transistor. The distribution of the stationary ions in these regions so that the potential barrier that appears can be attributed to the curve of figure 3 b. The height V_0 of this barrier cannot be exceeded by either the electrons or the holes of the N and P sections respectively, only with the small kinetic energy of the thermal vibration, and therefore any further diffusion stops as long as no external bias has been applied to the transistor.

Figure 1 PNP and NPN type contact transistors.

Figure 2 a) Construction model of a transistor b) Its potential barrier

The polarizations that we can apply are divided into three categories: a) normal operation b) cutoff c) saturation d) electrode reversal

A) Normal operation polarization

This is achieved if we polarize the emitter-base and reversely the base-collector (figure 3a). In this way, the emitter-base depletion region shrinks and the potential barrier is lowered from V_0 to $V_0 - V$ while the base-collector depletion region expands from V_0 to $V_0 + V_{CB}$ (figure 3b). Now the holes of the emitter P are repelled by the positive potential of the source V_{EB} and pass to the base N. There, because the base width is smaller, only a very small part of the holes will reconnect with electrons of the base, creating a very small base current, while the others, with the help of the negative electric field from the collector, will pass to it where they will be neutralized by the electrons pushed by the source V_{CB} and a large collector current is created. So if we apply Kirchhoff's first rule to the base, we will have:

$$I_{EBC}$$

Figure 3 The transistor in normal operating condition

b) Cut-off polarization

This is achieved if we reverse-bias the emitter-base and reverse-bias the base-collector. Then both depletion regions widen, resulting in no current flowing in no direction except for very small reverse currents I_{EO} and I_{CO} emitter - base and base - collector.

c) Saturation polarization

This is achieved if we properly bias the emitter - base and the base - collector. Then both stripping areas shrink resulting in movement holes towards the base from both the emitter and the collector resulting in the passage of holes from the emitter to the collector is prevented. Thus the barrier potential is completely eliminated and the transistor behaves like a common conductor and currents I should be limited with external resistors to avoid the transistor is damaged.

d) Reverse polarity of emitter-collector electrodes

In this case, the same applies as in normal operation, except that because Constructionally, the emitter and collector doping differs in terms of specifications, the transistor will operate with reduced efficiency and therefore such a connection is considered wrong and should be avoided.

BASIC TRANSISTOR CIRCUITS

The basic transistor circuits are:

- a. Common base (figure 4a)
- b. Common emitter (figure 4b)
- c. Common collector (figure 4c)

Figure 4 The basic circuits of a transistor with a) common base, b) common emitter and c) common collector.

In each of these we distinguish two loops that we call the input circuit (A) and the ~~output~~ circuit (B). In this exercise ~~we will deal~~ with the common emitter connection where we will plot the input, output and transfer characteristics (dynamic characteristic). We will also study the behavior of the transistor with the theory of quadrupoles and we will calculate its hybrid parameters in the equivalent circuit from the measurements that we will take for plotting the characteristics.

Feature engraving

To plot the characteristics of the transistor, we perform a connection (common emitter or base) and we take measurements. In figure (5b) we have a common emitter connection and the following characteristics.

A) Input characteristics

These are plotted if we keep the voltage V_{CE} constant and vary V_{BE} we observe a change in I (e.g. figure 5a for three different $V_{CE} = 1V, 10V, \text{ and } 20V$). The relationship given by this cluster is $V_{BE} / I_B \approx \text{constant}$. The characteristics of the forward biased base-emitter junction diode, I becomes large (in the order of μA) for a V_{BE} voltage of about 0.7 for Si transistor current or 0.3 for Ge transistors.

B) Output characteristics

These are drawn if we keep the current I_B constant and vary the

V_{CE} we observe the change of I_C (e.g. figure 5c for the following different $I_B = 0, 10 \mu A, 20 \mu A, 20 \mu A, 40 \mu A \text{ and } 50 \mu A$). The relationship given by this cluster is

$$I_C \propto (V_{CE}) \quad \text{for } I_B \text{ fixed}$$

Here we observe that initially for very small changes in V_{CE} I_C increases abruptly (interval when $V_{CB} > V_{EB}$ and therefore correct collector-base polarization) while, after almost stabilizes with a slight upward slope (interval that $V_{CB} < V_{EB}$ and hence reverse collector-base bias, hence widening of the stripping region collector - base, therefore limiting growth and controlling it through I_B).

Figure 5 a) Statistical characteristics of the input, b) the common emitter circuit and c) the statistical characteristics of the output.

In the following figure (7) we see the cluster of output features and the operating regions of the transistor depending on the bias we have.

Figure 6 Input circuit (a) and output (b) of the entire circuit.

Figure 7 The cut-off area and the saturation region should not be used in reinforcement.

C) Transfer characteristic

From the above we understand that what interests us most in a transistor is its behavior in amplifier circuits. This behavior shown by a curve connecting a magnitude of the input with a magnitude of the output. This curve is called the transfer characteristic and is shown in figure (8).

We observe that it results from the output characteristic curve if in

On the horizontal axis, we place the values of I and correspond them with the values of I .

c

In the output characteristic curve of figure (8) we observe another line called the load line and is the one that determines the characteristic transportation.

The load line results from the output circuit to which we connect a load resistor R_C , and is plotted in the output characteristic array, for a specific polarization as shown in the circuit in figure (6).

Applying Kirchoff's second law I have: $V_{CC} = I_C R_C + V_{CE}$ Expanding (1) for the two extreme situations:

A. the transistor in cutoff $I_B = 0$ and therefore $I_C = 0$

from (1) I will have $V_{CE} = V_{CC}$ (2)

B. Transistor in saturation $I_C = I_{C_{max}}$ and therefore $V_{CE} = 0V$

from (1) I will have $I_C = \frac{V_{CC}}{R_C}$ (3) $V_{CE} = 0V$

Equations (2) and (3) give two points on the axes I and V_{CC} at diagram of the output characteristics, with which I draw the load line. Each point of the load line can be chosen as the operating point Q. The choice of point is made with the base current I_B . Depending on the base current we choose and an output characteristic (from the swarm) and the intersection point of the characteristic with the The load line is the operating point.

A transistor, like any electrical element, consumes power during operation. its operation and an amount of power that is converted into heat and strains the transistor. So as not to exceed the consumption capabilities of the transistor and to use it in normal operating conditions, manufacturers also give us the maximum power that can be consumed at its collector. This is known as the maximum

collector loss and is given by the relationship:

$$P_{C_{max}} = I_C V_{CE}$$

The values that come out of this relationship represent an exaggeration in the figure. (8) shown below which divides the plane I_C , V_{CE} of the swarm of characteristics in two areas A($P_C < P_{C_{max}}$) and B($P_C > P_{C_{max}}$). As is evident, all the points in region B are forbidden and therefore the operating point of the transistor as well as the line should be in area A.

FIGURE 8 The dynamic characteristic and how to draw it**FIGURE 9** The curve

PC max and the limit position
of the straight line load

4. BEHAVIOR STUDY WITH THE THEORY OF QUADROPOL

We use quadrupole theory to derive the electrical equivalent of the transistor so that we can study its behavior in the various circuits. The theory of quadrupoles characterizes as a quadrupole any simple or composite that displays two input and two output terminals (figure 10). Ignoring the studying its internal composition, we arrive at conclusions with external

measurements made at its terminals, i.e. the currents

I_1 and the voltages V_1 ,

V_2 . Of these quantities, two can be considered independent variables while the two other functions of the previous ones. Thus, six (6) pairs of equations arise from which we will see the three that interest us.

$$\begin{aligned} V_1 &= f(I_1, I_2) \\ V_2 &= f(I_1, I_2) \end{aligned} \quad [P1]$$

$$\begin{aligned} I_1 &= f(V_1, V_2) \\ I_2 &= f(V_1, V_2) \end{aligned} \quad [P2]$$

FIGURE 10 A quadrupole
with the directions of currents and
of the trends that are considered
as positive.

$$\begin{aligned} V_1 &= f(I_1, I_2) \\ I_1 &= f(V_1, V_2) \end{aligned} \quad [P3]$$

That is, we can make various combinations of voltages and currents. so that by changing two of them, a third is changed. In P1 the change of $(I_1 \text{ or } I_2)$ changes V_1 or V_2 . Similarly the voltages (V_1, V_2) in P2 change the

I_1 and (I_1, V_2) in P3 change I_2 or V_2 . Always considering the quadrupole operation as linear and the transformations of

i voltages u_1, u_2 and low frequency i that they are small amplitude sinusoidal changes and

currents around the quiescent point of the DC bias of TR and V_1, V_2 ,

I_1, I_2 for the DC (static) operation of the quadrupole TR, while u_1, u_2, i its AC (dynamic) operation, as variations of the previous ones, i.e. where u_1, u_2 and so on, then y_1, y_2, y_3 are written respectively.

$$\begin{aligned} P1: \quad u_1 &= Z_{11} I_1 + Z_{12} I_2 \\ u_2 &= Z_{21} I_1 + Z_{22} I_2 \end{aligned} \quad (\text{where } Z \text{ parameters are impedances in } (\Omega))$$

$$P2: \quad I_1 = Y_{11} u_1 + Y_{12} u_2 \quad (\text{where parameters } Y \text{ are complex conductivities in } (S))$$

$$I_2 = Y_{21} u_1 + Y_{22} u_2$$

$$\begin{aligned} P3: \quad u_1 &= h_{11} I_1 + h_{12} I_2 \\ u_2 &= h_{21} I_1 + h_{22} I_2 \end{aligned} \quad (\text{where } h \text{ parameters hybrid (hybrid) the Greek word hybrid in English, i.e. mixed parameters in } (S),$$

(dimensionless or pure numbers) and $(\Omega)^{-1}$)

In each pair of the above equations the first represents the input and the second the output of the quadropole. But since the parameters Z and Y give us information only about the resistances and input-output conductivities of the quadropole we use the h which except

from the input resistance and output conductance they give information and the reasons reverse voltage and forward current transfer of the quadropole (active element TR).

If I consider the transistor as a quadropole with a common terminal for input and output the emitter (common emitter CE connection) as below shape, then the h are written as follows:

FIGURE 11 Transistor CE connection as a quadropole. in

$$\begin{aligned} &: U_{be} h_{ie} U_{re} \quad \text{this} \quad 4 \\ &: I_c h_{fc} I_{fd} \quad \text{oh} \quad \text{this} \end{aligned}$$

The first indices of the h parameters mean from the English meaning: i = input = input, f = forward = correct, o = output = exit and the second indices mean the transistor connection e = CE, so parameters h of the common connection transmitter.

From the functions P4 we can calculate the parameters h considering correspondingly zero some AC voltages or currents, e.g. $U_{ce} = 0$ (short circuit of the output) or we consider $V_{ce} = \text{constant value}$, e.g. if $I_b = 0$ (open-circuited input) or otherwise if $I_{\bar{B}}$ constant value, then $I_B = I_b = 0$.

This is how we define the h (hybrid parameters), which also have physical meaning:

Input impedance:	$h_{ie} = \frac{U_{be}}{I_b}$	in (Ω) for $U_{ce} = 0$ or $V_{ce} = \text{constant}$.
Reverse voltage transfer:	$h_{re} = \frac{U_{be}}{U_{ce}}$	(net number)/for $I_b = 0$ or $I_{\bar{B}} = \text{constant}$.
Correct current transfer:	$h_{fe} = \frac{I_c}{I_b}$	(net number)/ for $U_{ce} = 0$ or $V_{ce} = \text{constant}$.
Composite output conductivity:	$h_{oe} = \frac{I_c}{U_{ce}}$	in $(\frac{1}{\Omega})$ for $I_b = 0$ or $I_{\bar{B}} = \text{constant}$.

or output impedance

$$\sqrt{\frac{1}{h_{oh} I_c}}, \text{ in (Oh).}$$

From the previous definition of the hybrid parameters, the system of equations of P4 and the figure THE QUADRO-POLE TRANSISTOR, we can construct the hybrid equivalent circuit TR connection CE (for low frequencies), taking care that the OUTPUT equation is Kirchoff's 2nd law and INPUT 1th law:

The short circuit is due to a common output-input (CE).

FIGURE 2

Similarly, we can write the equations for the other connections and design the corresponding hybrid equivalent circuits TR using the method of quadrupole. So we have:

_____ $U_{ib} = h_{ib} I_b$ For CB: INPUT:
EXIT: $I_c = h_{cb} U_{cb}$

FIGURE 13 a1

FIGURE 13 a2

INPUT: U_{be} $i_{b,c}$ h_{re} For CC:
 i_{be} h_{fe} h_{oc} U_{ec} EXIT:

FIGURE 13 b1**FIGURE 13 b2**

Fig. 13 a1, b1 – The hybrid connections of a transistor and Fig. 13 a2, b2 – The hybrid their equivalents. The second index in the parameter symbols indicates the grounded quadrupole.

The h parameters can be measured in practice with instruments or estimated graphically from the clusters of its characteristics for the specific operating point ..., on the load line.

The calculation of the parameters h using the graphical method is shown in figure (14), solving equations P4 and P5 with respect to the parameter we want to calculate.

$$\begin{aligned}
 h_{re} &= \frac{V_{be}}{I_b} \text{ with } V_{ce} = \text{constant} \\
 h_{re} &= \frac{V}{V_{ce}} \text{ with } I_b \text{ fixed} \\
 h_{fe} &= \frac{I_c}{I_b} \text{ with } V_{ce} = \text{constant}
 \end{aligned}$$

$$h_{oh} = \frac{I_c}{V_{this}} \text{ with } I_b = \text{fixed}$$

In the figure below the calculation is done with common emitter connection but the procedure is the same for all basic connections.

$$h_{ie} = \frac{V_{be}}{I_{b \text{ } V_{ce} = V_{ce5}}}$$

$$= \frac{(0.65 - 0.55) \text{ V}}{(125 - 75) \text{ } \mu\text{A}} = 2$$

$$h_{fe} = \frac{I_c}{I_{b \text{ } V_{ce} = V_{ce5}}}$$

$$= \frac{(12.5 - 7.5) \text{ mA}}{(125 - 75) \text{ } \mu\text{A}} = 100$$

$$h_{re}/Q = \frac{V_{be}}{V_{Ib} / 100 \text{ } \bar{y}} = \frac{(0.75 \text{ } 0.6) \text{ } V}{(7.5 \text{ } 2.5)} = 300 \cdot 10^{-4}$$

$$h_{oh}/Q = \frac{I_c}{V_{Ib} / 100 \text{ } \bar{y}} = \frac{(10.5 \text{ } 9.5) \text{ } mA}{(7.5 \text{ } 2.5) \text{ } V} = 2 \cdot 10^{-4} \text{ Siemens}$$

Fig. 4 – Graphical estimation of the values of the h parameters in a transistor with connection common emitter.

LABORATORY PART

We connect the circuit of figure 15 to the breadboard.

FIGURE 15

FIGURE 16 – The legs of BC 107 when we look at it from below in the order E(Emitter), B(Base), C(Collector).

1a. We obtain the output characteristics for (i) $I_C (VCE)$ $I_{\bar{B}} 10 \mu A$, (ii) $I_{\bar{B}} 30 \mu A$, (iii) $I_{\bar{B}} 50 \mu A$, (iv) $I_{\bar{B}} 75 \mu A$, (v) $I_{\bar{B}} 100 \mu A$, (vi) $I_{\bar{B}} 130 \mu A$.

Vce (V)		0.1	0.3	0.6	0.9	1.5	2	3	4	5	7	9
Ib= 10 μA	IC(mA)											
Ib= 30 μA	IC(mA)											
Ib= 50 μA	IC(mA)											
Ib= 75 μA	IC(mA)											
Ib= 100 μA	IC(mA)											
Ib= 130 μA	IC(mA)											

Table 1

1b. From the above values in table 1, draw the output characteristics

$$I_{FC} \quad (V_{CE})$$

1c. Also from the above values of table 1 and on the graph (1b.) draw the straight line of load.

2a. We obtain the transfer characteristic I_{FC} for: $V_{CE} = 15V$

$I_b(\mu A)$	5	10	20	40	60	80				100	120	140	160	180	200		
$I_c(mA)$																	

Table 2

2b. From the values in table 2, draw the transfer characteristic I_{FC} (I_b)

3a. We obtain the input characteristics V_{be} I_{FC} for: $V_{CE} = 0V$

$V_{be}(V)$	0.2	0.3	0.4	0.5	0.55	0.58	0.6	0.63	0.65	0.68	0.7	0.72				
$I_b(\mu A)$																

Table 3a

3b. We obtain the input characteristics V_{be} I_{FC} for: $V_{CE} = 7.5V$

$V_{be}(V)$	0.2	0.3	0.4	0.5	0.55	0.58	0.6	0.63	0.65	0.68	0.7	0.72				
$I_b(\mu A)$																

Table 3b

3c. Draw the input characteristics and 3b. I_{FC} from the above values of table 3a

QUESTIONS

1. Based on the construction and operation of the transistor, explain the form of the output, input and transfer characteristics (dynamic characteristic).

2. What is the usefulness of output characteristics and what does the load line determine?

3. What is β (h_{fe}) and what does it determine for each transistor? In the circuit of Fig. 15 the I_{c1} is 15mA and I_{b1} is 30 μA , calculate the amplification factor β or β of circuit.

4. a) What is the need for the existence and use of hybrid parameters?

b) Calculate from the measurements you took and from the graphs that you designed, all the hybrid parameters.

5. a) Complete the table below. The following applies:

(for $P=300\text{mW}$)

$V_{ce}(\text{V})$	1.5	2	3	4	5	7	9
$I_c(\text{mA})$							

b) draw the total useful power curve for:

EXERCISE 9^{OR}
LOGIC GATES
WITH DIODES AND TRANSISTORS

Make the circuit of Figure 1.

Figure 1

Short-circuit point A (diode cathode) with +5V. On the voltmeter obviously we will measure +5V since the diode does not conduct (There is no difference potential at its ends) and no current flows through the circuit.

Now disconnect point A from +5V and connect it to ground (0V). The diode becomes conductive and at its ends we will measure 0.3V if it is Germanium and 0.6V if it is Silicon.

Now make the circuit of figure 2 which is also the circuit of a AND gate with two inputs A and B.